

Online appendix for “No Country for Old Men: Aging, Innovation, and Dynamic Inefficiency”

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A Proof of Proposition 5 (Structural decomposition)

Proof. We prove cases (i)–(iii) and provide a proof sketch for case (iv). Throughout, we work at an interior CE that satisfies the model’s first-order conditions of Section 4 of the main text, with within-period CES bundling and constant within-type elasticity $\varepsilon > 1$ (so price indices satisfy $P_s = Q_s^{1/(1-\varepsilon)} \mu w$ in symmetric equilibrium, $\mu = \varepsilon/(\varepsilon - 1)$)).

Case (i): single-technology economy. With a single technology type the composition variable Q_n is degenerate ($Q_n \equiv 1$ or $Q_n \equiv 0$), so the wedge $Q_n^{CE} - Q_n^*$ is undefined.

Case (ii): homogeneous demand implies $Q_n^{CE} = Q_n^$.* Suppose all consumer types have identical preferences and identical effective wealth, so the demand side collapses to a representative consumer with period utility $u = \theta \ln c_n + (1 - \theta) \ln c_r$ and income m . The indirect utility function is

$$V(P_n, P_r, m) = \ln m - \theta \ln P_n - (1 - \theta) \ln P_r + \text{const.} \quad (1)$$

Step 1: the variety gradient. Using the symmetric-equilibrium price indices $P_s = Q_s^{1/(1-\varepsilon)} \mu w$ (so $\partial P_n / \partial Q_n = -P_n / [(\varepsilon - 1)Q_n] < 0$ and $\partial P_r / \partial Q_n = +P_r / [(\varepsilon - 1)(1 - Q_n)] > 0$),

$$-\theta \ln P_n - (1 - \theta) \ln P_r = \text{const} + \frac{1}{\varepsilon - 1} \left[\theta \ln Q_n + (1 - \theta) \ln(1 - Q_n) \right], \quad (2)$$

so the direct (price-index) gradient of utility with respect to composition is

$$\left. \frac{\partial V}{\partial Q_n} \right|_{m \text{ fixed}} = \frac{1}{\varepsilon - 1} \left[\frac{\theta}{Q_n} - \frac{1 - \theta}{1 - Q_n} \right]. \quad (3)$$

Step 2: income accounting. With Cobb–Douglas spending across nests, expenditure on nest s is $E_n = \theta m$, $E_r = (1 - \theta)m$. Each of the Q_s symmetric firms in nest s earns revenue E_s/Q_s and, at constant markup $\mu = \varepsilon/(\varepsilon - 1)$ over marginal cost w , a profit equal to the share $1/\varepsilon$ of its revenue. Aggregate operating profit is therefore

$$\Pi = \frac{E_n + E_r}{\varepsilon} = \frac{m}{\varepsilon}, \quad (4)$$

independently of the composition Q_n . With aggregate innovation outlay $C_\Phi = Q_n^- \Phi(e_n) + (1 - Q_n^-) \Phi(e_r)$ paid by the inherited-type masses (within-period realization timing), dividends are $D = \Pi - C_\Phi$ and the representative consumer’s budget $m = w + D$ gives

$$m = \frac{\varepsilon}{\varepsilon - 1} (w - C_\Phi). \quad (5)$$

Composition affects income only through the resource cost of innovation, and the markup multiplier $\varepsilon/(\varepsilon - 1)$ scales the social cost of diverting labor from production to R&D.

Step 3: the planner's first-order condition. The planner chooses efforts (e_n, e_r) given the inherited composition Q_n^- , with realized composition $Q_n = p(n, e_n) Q_n^- + p(r, e_r) (1 - Q_n^-)$, so $\partial Q_n / \partial e_s = m_s p_e(s, e_s)$ where m_s is the mass of inherited type s . Combining (3) and (5), the money-metric welfare gradient ($\lambda = \partial V / \partial m = 1/m$) is

$$\frac{1}{\lambda} \frac{dV}{de_s} = -\frac{\varepsilon}{\varepsilon-1} m_s \Phi'(e_s) + \frac{m}{\varepsilon-1} \left[\frac{\theta}{Q_n} - \frac{1-\theta}{1-Q_n} \right] m_s p_e(s, e_s), \quad (6)$$

and the planner's interior optimum satisfies

$$\Phi'(e_s) = p_e(s, e_s) \cdot \frac{m}{\varepsilon} \left[\frac{\theta}{Q_n} - \frac{1-\theta}{1-Q_n} \right]. \quad (7)$$

Step 4: the firm's first-order condition. Per-firm operating profit in nest s is $\pi_s = E_s/(\varepsilon Q_s)$, so the profit gap is

$$\Delta\pi = \pi_n - \pi_r = \frac{m}{\varepsilon} \left[\frac{\theta}{Q_n} - \frac{1-\theta}{1-Q_n} \right], \quad (8)$$

and the firm's static first-order condition $\Phi'(e_s) = p_e(s, e_s) \Delta\pi$ is *identical* to the planner's condition (7).

The result reflects the exact cancellation of two markup distortions. The consumer-surplus gradient of composition (3) exceeds the private profit gap (8) by the factor $\mu = \varepsilon/(\varepsilon - 1)$: consumers value variety at marginal utility, while firms appropriate only the profit share $1/\varepsilon$ of revenue. But by (5) the social cost of innovation exceeds the firm's private cost by the same factor μ : a unit of labor diverted from production to R&D reduces markup-inclusive final spending by μ . The two factors cancel, the first-order conditions coincide at every state, and $Q_n^{CE} = Q_n^*$. This is the composition analogue of the constrained efficiency of the CES monopolistic-competition benchmark (Dixit and Stiglitz, 1977; Dhingra and Morrow, 2019): with a common, constant within-nest elasticity and Cobb–Douglas spending across nests, the market's composition choice is undistorted for a representative consumer.

Dynamic extension. With the continuation term, the firm's condition is $\Phi'(e_s) = p_e(s, e_s) (\Delta\pi + q \Delta J)$, with firm values satisfying the fixed point

$$J_s = d_s + q (p_{ss'} J_n + (1 - p_{ss'}) J_r). \quad (9)$$

Under a representative consumer, $q = \beta\lambda'/\lambda$ is the consumer's own intertemporal marginal rate of substitution, so the market value gap $q \Delta J$ prices the future composition consequences at exactly the consumer's valuation, and the factor- μ cancellation

of the static argument applies period by period, so the wedge is zero at every horizon.

Case (iii): disjoint inputs. Suppose factor A is used exclusively in the production of new-technology goods and factor B exclusively in retro-technology goods. The labor-market clearing conditions decouple by technology:

$$Q_n \cdot \ell_A^{\text{firm}} = L_A, \quad (1 - Q_n) \cdot \ell_B^{\text{firm}} = L_B, \quad (10)$$

where L_A, L_B are the exogenous endowments of factors A, B . Since ℓ_s^{firm} is determined by the firm's cost-minimizing labor demand at given prices, Q_n is pinned by the relative factor endowments L_A/L_B and is no longer a free firm choice. The constrained-efficient planner faces the same constraint, so Q_n^* equals Q_n^{CE} trivially and the wedge is zero.

Case (iv): differentiated inputs under (B) — proof sketch. Suppose both technology types use both factors A and B but in different intensities (new-tech A -intensive, retro B -intensive), with inelastic endowments L_A, L_B . Factor-market clearing then makes equilibrium factor prices functions of the composition, $w_A(Q_n)$ and $w_B(Q_n)$, with $w'_A(Q_n) > 0$ and $w'_B(Q_n) < 0$: a larger new-technology sector bids up the price of the factor it uses intensively.

Atomistic firms take factor prices as given, so the factor-price response enters the *equilibrium mapping* from composition to profits rather than any individual firm's first-order condition. Writing per-firm profits at the market-clearing factor prices as $\pi_s(Q_n) \equiv \pi_s(w_A(Q_n), w_B(Q_n); Q_n)$, the equilibrium profit gap

$$\Delta\pi(Q_n) = \pi_n(Q_n) - \pi_r(Q_n) \quad (11)$$

is more steeply declining in Q_n than under shared inputs: as Q_n rises, w_A rises and w_B falls, compressing π_n relative to π_r over and above the demand-side compression operating through the price indices. The firm's first-order condition remains $\Phi'(e_s) = p_e(s, e_s) [\Delta\pi(Q_n) + q \Delta J(Q_n)]$, evaluated at the equilibrium objects. This general-equilibrium feedback is the precise sense in which factor prices “partially internalize the supply side”: any given gap between private and social composition incentives translates into a smaller equilibrium deviation of Q_n , because the private incentive itself responds more steeply to Q_n , attenuating the wedge.

The wedge does not vanish, however, because the demand-side discrepancy is untouched by factor prices. Under (B), consumer types have different marginal utilities of income λ_i , and the planner's money-metric valuation of a composition change weights each type's price-index exposure by λ_i (Roy's identity type by type, $\partial V_i / \partial P_s = -\lambda_i c_{i,s}$, with $\partial P_n / \partial Q_n = -P_n / [(\varepsilon - 1)Q_n]$ and $\partial P_r / \partial Q_n = +P_r / [(\varepsilon - 1)(1 - Q_n)]$). The private profit gap aggregates the same exposures by

expenditure, not by marginal utility. The difference between the two aggregations,

$$\sum_i \left(\frac{\lambda_i}{\bar{\lambda}} - 1 \right) \left[\frac{c_{i,n} P_n}{(\varepsilon - 1) Q_n} - \frac{c_{i,r} P_r}{(\varepsilon - 1) (1 - Q_n)} \right], \quad (12)$$

where $\bar{\lambda}$ is the planner's money-metric numeraire, is generically nonzero whenever λ_i varies across types and types differ in their exposure to the two price indices — which is precisely condition (B). No factor-price adjustment enters (12); hence the wedge generically survives differentiated inputs.

If, in addition, demand is homogenized ((B) fails), then $\lambda_i = \bar{\lambda}$ for all i and (12) vanishes identically. The cancellation argument of Case (ii) then extends, with factor income $w_A L_A + w_B L_B$ replacing w in the income accounting (5), and the wedge is zero. We do not attempt a closed-form magnitude for the attenuation in the differentiated-inputs case; the claim that the wedge *shrinks* is the qualitative general-equilibrium compression of $\Delta\pi(Q_n)$ described above, and we flag it as such in the main text. ■

B A sufficient statistic for the wealth wedge

This section derives the approximate sufficient statistic of Proposition 6 in the main text: in the static economy with $\psi = 0$ and log Cobb–Douglas preferences,

$$1 + \tau_0 = \frac{m}{2} (s_Y \lambda_Y + s_O \lambda_O), \quad (13)$$

where $\lambda_i = 1/W_i$, $m = W_Y + W_O$, and (s_Y, s_O) is the incidence of a marginal innovation outlay on the two cohorts' incomes.

Setup. Two cohorts with incomes W_Y, W_O and identical log Cobb–Douglas preferences ($\psi = 0$, so both spend share θ on new-technology goods). Utilitarian welfare is

$$W = \ln W_Y + \ln W_O - 2\theta \ln P_n - 2(1 - \theta) \ln P_r + \text{const.} \quad (14)$$

Variety gradient. By the same computation as in (2), the price-index terms contribute

$$\left. \frac{\partial W}{\partial Q_n} \right|_{\text{incomes fixed}} = \frac{2}{\varepsilon - 1} g(Q_n), \quad g(Q) \equiv \frac{\theta}{Q} - \frac{1 - \theta}{1 - Q}, \quad (15)$$

the factor of two reflecting that both cohorts enjoy the variety effect.

Income accounting with incidence. As in (4)–(5), aggregate operating profit is the constant share m/ε of spending, so $D = w/(\varepsilon - 1) - \mu C_\Phi$ with $\mu = \varepsilon/(\varepsilon - 1)$: a marginal innovation outlay $\Phi'(e_s)$ by the mass- m_s inherited type reduces aggregate income by $\mu m_s \Phi'(e_s)$. Let the reduction fall on the two cohorts in proportions (s_Y, s_O) , $s_Y + s_O = 1$. In the model the split is determined endogenously by the asset structure: outlays reduce dividends and the cum-dividend firm value the old

sell, with part of the reduction capitalized into the price the young pay for shares. The derivation treats (s_Y, s_O) as a summary parameter.

Planner versus market. The planner's first-order condition for effort e_s balances the variety gradient against the incidence-weighted income loss:

$$(s_Y \lambda_Y + s_O \lambda_O) \mu \Phi'(e_s) = \frac{2}{\varepsilon - 1} g(Q_n) p_e(s, e_s), \quad (16)$$

that is, using $(\varepsilon - 1) \mu = \varepsilon$,

$$\Phi'(e_s) = p_e(s, e_s) \cdot g(Q_n) \cdot \frac{2}{\varepsilon (s_Y \lambda_Y + s_O \lambda_O)}. \quad (17)$$

The market's first-order condition uses the private profit gap. With both cohorts spending share θ on new technology, $E_n = \theta m$ and $\Delta\pi = (m/\varepsilon) g(Q_n)$, so

$$\Phi'(e_s) = p_e(s, e_s) \cdot g(Q_n) \cdot \frac{m}{\varepsilon}. \quad (18)$$

The corrective tax that aligns (18) with (17) is the ratio of the two marginal benefits:

$$1 + \tau_0 = \frac{m/\varepsilon}{2/[\varepsilon (s_Y \lambda_Y + s_O \lambda_O)]} = \frac{m}{2} (s_Y \lambda_Y + s_O \lambda_O), \quad (19)$$

which is (13).

Limiting cases and the criterion. If $\lambda_Y = \lambda_O = \lambda$ then $W_Y = W_O = m/2$ and $1 + \tau_0 = (m/2)\lambda = 1$: the representative-agent benchmark of case (ii), recovered exactly. The wedge also vanishes at the knife-edge incidence $s_O^* = (\bar{\lambda} - \lambda_Y)/(\lambda_O - \lambda_Y)$ with $\bar{\lambda} = 2/m$, where the incidence-weighted marginal utility equals the population average. Under the generation-discounted criterion $u_Y + \beta u_O$, the welfare function weights the old's log income and variety terms by β ; repeating the derivation gives

$$1 + \tau_0 = \frac{m (s_Y \lambda_Y + \beta s_O \lambda_O)}{1 + \beta}. \quad (20)$$

Verification. At the exact $\psi = 0$ stationary equilibrium of the calibrated model, $W_Y = 0.735$, $W_O = 0.285$, $m = 1.020$, so $\lambda_O/\lambda_Y = 2.58$. Formula (13) brackets the wedge between $\tau_0 = 24\%$ ($s_O = \frac{1}{2}$) and $\tau_0 = 79\%$ ($s_O = 1$); the model's $\tau_0 = 55\%$ corresponds to $s_O \approx 0.78$. Formula (20) at the same incidence predicts $\tau_0 \approx -3.5\%$ under the generation-discounted criterion, matching the sign and order of magnitude of the full model's -15% (which additionally includes the adoption-cost channel and dynamic effects). The derivation is a static approximation: it abstracts from the continuation values, the equal-marginal-cost linkage across firm types, and the endogeneity of (s_Y, s_O) , which is why it is stated as an approximate sufficient statistic rather than

an exact decomposition.

C On-path optimality of the trajectory-iteration fixed point

This section proves the on-path optimality claim of Proposition 8 in the main text: any fixed point of the iterative best-response algorithm satisfies the Bellman equation of the dynamic constrained-efficient problem at every state realized along the path, and is therefore the on-path realization of a Markov-perfect policy. Convergence of the iteration itself is a numerical property, not claimed in general; at the calibration the predicted-versus-realized error contracts at a rate of roughly 0.4–0.6 per outer iteration and falls below 10^{-6} .

Proof. Let $\{Q_n^*(t)\}_{t=1}^T$ be a fixed point of the iteration: the trajectory produced by the period-by-period best responses coincides with the trajectory on which the rational-expectations aggregates $\{V(t), \lambda_O(t)\}$ of the backward pass were computed. Three observations complete the argument.

(1) *Consistency of anchors.* At a fixed point, the continuation aggregates entering each period’s local problem are generated by the fixed-point trajectory itself: the anchored flows of Proposition 7 coincide with the equilibrium flows along the realized path.

(2) *One-shot deviations are unprofitable.* By construction of the forward pass, $Q_n^*(t)$ maximizes the local objective $u(t) + u(t+1)$ given the inherited state $Q_n^*(t-1)$, the future choice $Q_n^*(t+1)$, and the fixed-point aggregates. By Proposition 7, this local objective differs from the full continuation objective $\sum_{s \geq t} u(s)$ by terms independent of $Q_n(t)$. Hence no one-shot deviation at any on-path state raises the planner’s objective.

(3) *One-shot deviation principle.* The horizon is finite, the feasible set for $Q_n(t)$ is the compact interval implied by the law of motion, and at the calibration the local objective is continuous and strictly concave on that interval at every visited state. For finite-horizon problems, unimprovability by one-shot deviations implies Bellman optimality at every reachable state, by backward induction over $t = T, \dots, 1$. Therefore $\{Q_n^*(t)\}$ satisfies the Bellman equation at every state realized along the path, which is the defining property of a Markov-perfect policy restricted to the realized trajectory. ■

Two scoping remarks are in order. First, the argument establishes optimality *on the realized path only*: states off the path are never visited by the algorithm, and the policy is not computed there. The main text states this scope explicitly. Second, existence and uniqueness of the fixed point are not claimed in general; both are verified numerically at the calibration, including a multi-start check from six diverse initial trajectories.

References

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- Dixit, Avinash K. and Joseph E. Stiglitz. 1977. “Monopolistic Competition and Optimum Product Diversity.” *American Economic Review* 67 (3):297–308.